

- 1 The drag force F_D acting on a sphere falling through a liquid is given by

$$F_D = 6\pi\eta rv$$

where r is the radius of the sphere,
 v is the speed of the sphere in the liquid and
 η is a property of the liquid called the viscosity.

- (a) Show that the SI base units of viscosity are $\text{kg m}^{-1} \text{s}^{-1}$.

[2]

- (b) The sphere has a radius of 3.0 cm and is falling vertically downwards at a terminal velocity of 2.0 m s^{-1} through the liquid. The drag force acting on the sphere is 0.096 N.

Calculate the viscosity of the liquid.

viscosity = $\text{kg m}^{-1} \text{s}^{-1}$ [2]

- (c) The sphere is shown in Fig. 1.1.

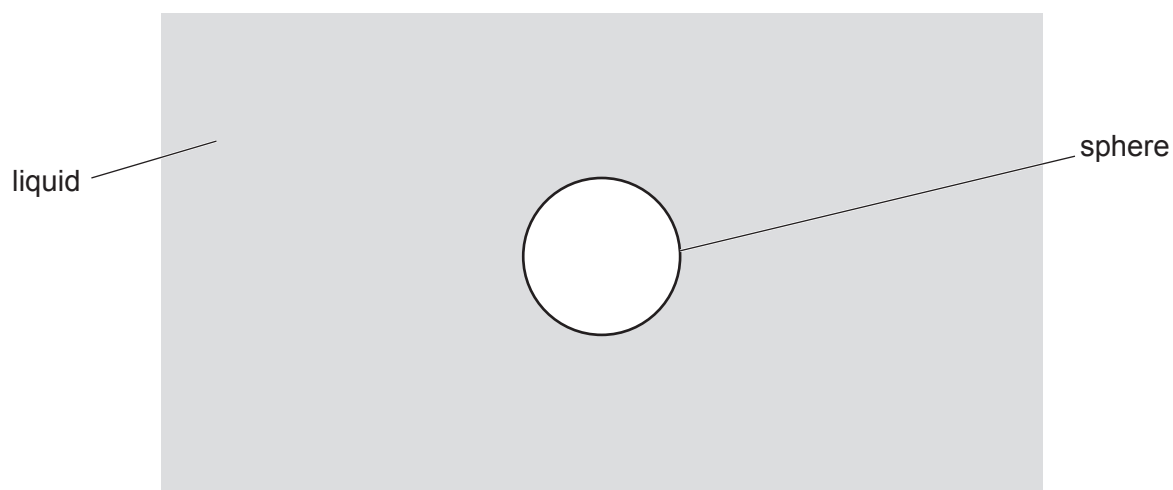


Fig. 1.1

On Fig. 1.1, draw and label arrows to represent the directions of the **three** forces acting on the sphere as it falls at terminal velocity. [2]

- (d) (i) The density of the liquid is 920 kg m^{-3} .

Show that the upthrust acting on the sphere is 1.0 N .

[2]

- (ii) Calculate the mass of the sphere.

mass = kg [2]

[Total: 10]